Corporate Editing and Collective Intelligence in OpenStreetMap: A Long-Term Analysis of Southeast Asian Case Studies

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As the most successful example of volunteered geographic information (VGI) [1], OpenStreetMap (OSM) has attracted considerable attention from various actors and communities [2], including corporations such as Meta, Amazon, and Uber [3]. Corporate editing (CE) teams can deploy coordinated, semi-automated, and AI-assisted workflows at scale [4,5]. This raises questions about changing modes of collaborative mapping when various actors participate under different institutional logics [2]. The literature documents distinct corporate contribution patterns [3,4], co-editing between paid and volunteer mappers [5], and CE impacts, e.g. on data quality [6]. Yet the empirical evidence remains fragmented, focusing on single outcomes or short observation windows, and leaving open the evaluative question of how CE changes OSM as a collective production system across multiple dimensions at once.

We address this gap through the collective intelligence (CI) framework. CI has previously been used to conceptualize and analyze OSM-related mapping systems, considering contributor composition, collective action, and evidence of intelligent outcomes together rather than as separate dimensions of data production [7,8]. CI is associated with independence of contributors, diversity of opinion, decentralization of decision-making, and effective aggregation as enabling conditions [9]. These correspond directly to CE: a CE campaign may improve aggregation by adding missing entities, while simultaneously reducing diversity or concentrating practice around a narrower set of workflows. The CI framing therefore lets us interpret improving and deteriorating outcomes together, asking whether CE sustains, erodes, or transforms OSM's collective intelligence.

We operationalize CI through measures that reflect CI's four conditions. Data quality proxies aggregation, assuming that a well-functioning mechanism results in better data. It is measured through entity count (completeness), geometric complexity (building-area-to-convex-hull-area ratio, or road node density), and tag richness. Diversity is measured through Shannon's diversity index applied to tags, as this reflects a diversity of semantic operations, and through innovation, i.e. new semantic notions: unique tags, new tag keys, new key-value

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pairs, and, of these pairs, the number propagating to 1% or 5% of all entities. Shannon evenness proxies decentralization and independence, with low values signaling convergence toward homogeneous tagging.

Our implementation covers five countries - Thailand (TH), Malaysia (MY), Myanmar, the Philippines, and Papua New Guinea - over 2015-2025. TH and MY were the focal countries of the early Facebook AI-Assisted Road Tracing project [10], and the other countries are used as control case studies. TH and MY are also analyzed at the province and state level (77 provinces, 16 states). CI measures are calculated from annual snapshots, collected using the ohsome API [11], representing the OSM database state in each region on 1 January of each year, and relating to buildings (*building=**) and roads (*highway=**). The code for this is available at https://github.com/Geo-Cultural-Informatics-Lab/OSM-report.

Our treatment variable is the log number of CE contributions, derived using organized-editing hashtags to identify CE contributions and classify them by edit type. The effect of the treatment is assessed in relation to each CI measure as well as to a composite CI score. This is derived using the Entropy Weight Method (EWM), which assigns each indicator an objective weight reflecting how strongly it discriminates between observations, and the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), which scores each unit by its closeness to an ideal-best profile.

While the country-level analysis is descriptive, the province-level analysis estimates a distributed-lag panel model: immediate and delayed effects on each CI dimension are regressed on CE contributions at lags of zero to three years; province and year fixed effects are added as controls; and a lagged outcome captures persistence. Treatment is split into Facebook, Grab, and outside-wave streams to separate actors (see details on waves below). We also add contemporaneous and one-year-lagged CE volumes among neighboring provinces to capture spatial and spatio-temporal spillovers. Outcomes are standardized so effects are comparable, and standard errors are clustered by province, with Driscoll-Kraay standard errors. The fully reproducible analysis is available at https://github.com/Geo-Cultural-Informatics-Lab/ce-impacts-ci.

We find that CE arrives in distinct regimes, with different timings marking two waves: the first wave, led by Facebook, occurs in 2017-2019, while increasing engagement from Grab since 2023 marks the second. Facebook's wave is much larger in total contribution volume and strongly creation-heavy, in contrast to Grab's orientation toward refinement and its focus on geometry and tag modifications (Fig. 1a). National-scale results show that CE does not necessarily mean higher CI. While permanent improvement is registered for some measures after CE engagement, e.g. highway geometric complexity after the first wave in both MY and TH, others experience a decrease, either temporary (e.g. unique highway tag counts in MY) or lasting (e.g. building tag evenness or diversity in TH). Overall impacts on CI seem to get absorbed for Thailand while a lasting effect is observed in MY (Figure 1b). However, it does not seem that CE consistently produces higher CI levels in comparison to the control countries: across many measures, TH's and MY's trajectories do not meaningfully differentiate from those of the controls. The high performance of the Philippines across measures, with its active and diverse OSM community [2], suggests that diversity and commitment can produce outcomes comparable, or even superior, to those derived from technologically and financially supported CE.

Province/state-scale analysis exposes the contextual nature of CE impacts. In TH, the Facebook wave correlates with expected road outcomes, i.e. an increase in highway entity counts and geometric complexity accompanied by decreasing evenness and diversity, producing a fuller and more homogeneous data landscape (Fig. 1c). The negative coefficients for highway counts and complexity in MY are thus surprising, showing how outcomes vary when the same process is implemented in different locations. The fact that TH shows many more significant coefficients than MY further emphasizes this. Variation also emerges across firms, with Facebook's activity correlated with more indices and at times contradicting Grab's impacts, e.g. its negative correlation with building tag evenness versus Grab's positive one. Together, these indicate that CE is not monolithic; its nature changes by location and by who carries it out (as already seen in Fig. 1a).

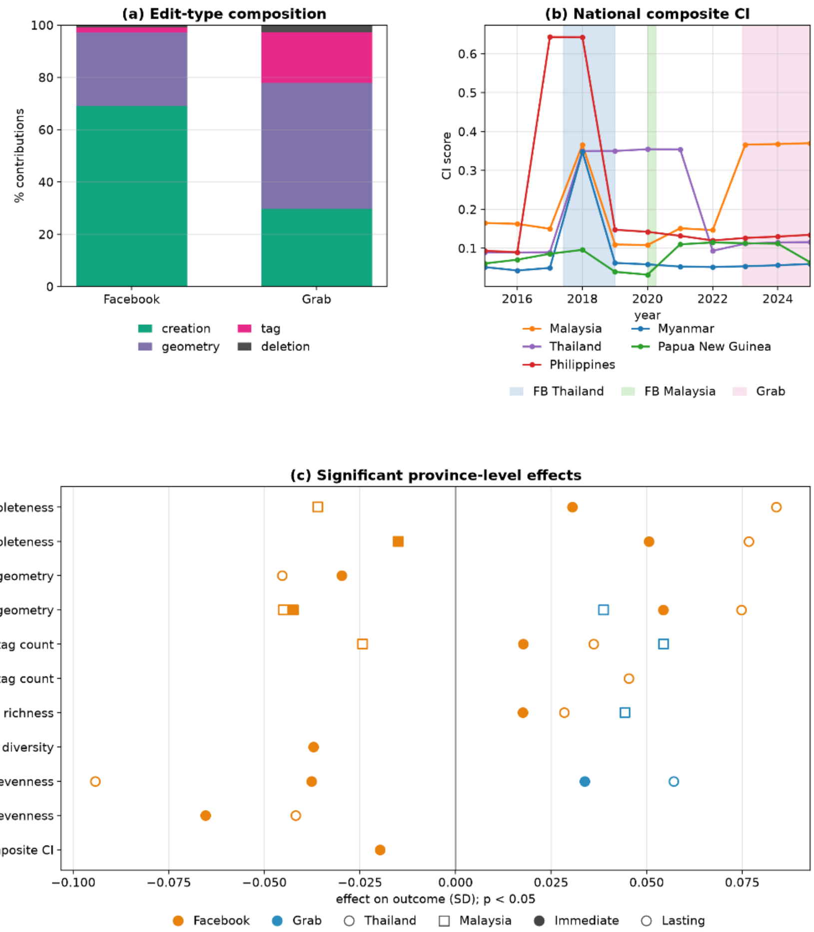

Figure 1. Selected analysis results: (a) Edit-type shares, by corporate actor; (b) composite CI scores per country over time; (c) estimation results - standardized coefficients significant at 95% confidence, by country, corporate actor, and duration of impact.

Immediate and lasting effects within distinct country-company combinations seem to show more homogeneity, always having the same sign when significant. Temporal processes may still vary: insignificant or lower lasting impacts signify absorption of the effect, while cases where the lasting coefficient exceeds the immediate one mean that the effect accentuates and path dependence emerges. Fortunately, this is positive in most cases. Spatio-temporal processes are also noteworthy: most significant spatial coefficients in both MY and TH relate to highways, which is logical as CE spreads across space; in TH, however, building complexity and unique tag counts also show spatial dependence. More importantly, most spatio-temporal effects in TH relate to buildings. This, together with significant patterns seen in Fig. 1c for buildings, suggests a shift in mapping activity, especially in TH, toward new complementary spatial and thematic focuses.

While these results may be unique to Southeast Asia and the early intervention by Facebook, they still portray a nuanced picture of CE activity and its impacts. Most of all, CE emerges not as a binary bad/good intervention, but as a contextual process and outcome. This contextual nature means that both its direct and lasting effects are not entirely predictable, at times not even producing the expected results given their specific focus. Furthermore, our results give the impression of resilience, with national-level effects being absorbed for many metrics. Lasting effects are registered at the province scale, but these are also absorbed at times and, when not, are frequently positive. Finally, spatio-temporal patterns suggest a dynamic response by the larger community, leading to complementary mapping patterns. This is not to say that CE does only good for the project and the data, as some effects are negative. Instead, it shows corporations as another set of actors within the OSM community landscape, one that despite its capabilities does not necessarily overpower others but still has the capacity to change dynamics. Corporations should therefore not be portrayed as external to the project: like humanitarian institutions, local chapters, governmental agencies, and others, they play a delicate role in making OSM and should be invited into open discussion with all other actors.

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